Ride Safety: Designing a Safer Ride-Sharing Experience for Women in Dhaka

Md. Rafiul Islam Ridoy
ID: 0112310032
Dept. of CSE
United International University
Dhaka, Bangladesh

Robayet Rowshan Siam
ID: 0112310415
Dept. of CSE
United International University
Dhaka, Bangladesh

A.B.M. Nurul Huda
ID: 0112310056
Dept. of CSE
United International University
Dhaka, Bangladesh

Samiun Alim Khan
ID: 0112230228
Dept. of CSE
United International University
Dhaka, Bangladesh

Israt Jahan
ID: 0112230056
Dept. of CSE
United International University
Dhaka, Bangladesh

Md. Sameem Kuddus Nehal
ID: 0112230186
Dept. of CSE
United International University
Dhaka, Bangladesh

***Abstract*—Women in Dhaka face persistent safety concerns while using ride-sharing services such as Uber, Pathao, and inDrive, yet the safety controls provided by these platforms are often hidden, reactive, and require multiple taps during moments of distress. This paper presents NariSafe, an HCI-driven prototype designed through a user-centered approach to improve the safety experience of female ride-sharing users in Dhaka. We combined a survey of 16 female riders, two in-depth semi-structured interviews, and affinity diagramming to identify six themes: Safety Concerns, Verification, Trusted Support, Emergency Alerts, Awareness Gap, and Privacy & Trust. These themes informed a five-feature prototype: one-tap live location, verified driver and vehicle card, unexpected route alert, SOS to trusted contact, and private incident reporting. Our findings show that visibility of safety controls, fast trusted-person support, and proactive route monitoring are the most requested design directions. We contribute design implications for gender-sensitive ride-sharing safety in the Global South.**

***Index Terms*—women's safety, ride-sharing, HCI, Dhaka, survey, thematic analysis, user-centered design, feminist HCI**

I. Introduction

A. Background

Women in Dhaka face safety risks across the full ride-sharing journey—before accepting a ride, during the trip, and after arrival. Ride-sharing platforms such as Uber, Pathao, and inDrive are widely used, but their safety controls are often hidden inside menus, slow to access, and reactive rather than preventive. Prior work has shown that 94% of women commuting by public transport in Dhaka experience some form of harassment [1], and studies on ride-sharing have identified driver behavior, route uncertainty, and traveling alone at night as recurring concerns [1], [6].

B. Motivation

Existing safety applications provide emergency alerts and GPS tracking, but they largely rely on the rider noticing danger first and then acting manually [8], [10]. In the Global South, unreliable internet, family oversight, and cultural norms around modesty ("Shotitwo") further shape what safety design is acceptable [3]. A previous study of Cape Town ride-share users found that personal safety perceptions differ sharply by gender and were affected by the Covid-19 pandemic [5]. This motivates us to investigate how safety design can be made proactive, culturally aligned, and offline-capable.

C. Objective

This paper addresses the gap between the safety features that exist and the safety that women actually feel. We conducted a survey of 16 female ride-sharing users, two in-depth semi-structured interviews, and an affinity diagramming session to derive six themes. Based on these themes, we designed and evaluated NariSafe, a prototype with five features: one-tap live location, verified driver and vehicle card, unexpected route alert, SOS to trusted contact, and private incident reporting.

D. Research Questions

This study addresses the following research questions:

- **RQ1:** What safety concerns do women face when using ride-sharing services in Dhaka?
- **RQ2:** How usable are the existing safety features in popular ride-sharing applications?
- **RQ3:** How can a user-centered mobile application improve the safety experience of female commuters in Dhaka?

Our contributions are: (1) empirical evidence on women's ride-sharing safety concerns in Dhaka, (2) a thematic analysis that translates these concerns into design requirements, and (3) a prototype that extends safety from basic ride support to proactive, offline-capable, evidence-aware protection.

II. Literature Review

A. Women's Safety in Public Transport in the Global South

Women in the Global South face disproportionate safety risks while commuting. Studies in India, Bangladesh, and

TABLE I: Existing safety applications and design characteristics

App	Key Features	Limitations
bSafe	SOS, location sharing	Internet required
Raksha	Alerts, GPS	No offline
Himmat	SOS, police link	Slow response
Shake2Safety	Shake SOS	False triggers
SafeStreet	Crowd-sourced map	No real-time alert
Navi.io	Safe route, reports	Internet required
Protibadi	Story reporting	No emergency response
NariSafe	Offline SOS, vehicle check, community	Prototype only

Pakistan report high rates of harassment and assault on public transport [3], [4], [8]. In Bangladesh, cultural norms around modesty and family reputation often prevent women from reporting incidents [3]. Asha et al. found that Bangladeshi women prefer preventive measures that avoid harassment altogether over reactive justice [3]. This preference has a direct design implication: a safety app should help users avoid the situation in the first place, not only react after it has occurred.

Karusala and Kumar examined India's national mandate to add panic buttons to every mobile phone and found that the intervention was ineffective because of poor police response, distrust in authorities, and the reactive nature of the feature [8]. Women reported that they would rather contact a family member than the police.

B. Ride-Sharing Safety in Dhaka

Abdal et al. surveyed 150 female ride-sharing users in Dhaka and found that verbal abuse was the most common form of harassment, followed by gestural and physical abuse [1]. Younger women (18–25) and those traveling alone or in uncrowded areas were most vulnerable. Meshram et al. found similar patterns in India, where women felt less safe sharing rides with unknown males or traveling at night [9].

Hu and Yang further showed that safety expectation, platform trust, and perceived environment shape female passengers' safety perception in ride-hailing [6], [11]. Yang et al. used structural equation modeling on 596 Chinese female passengers and found that safety expectation and platform trust are the strongest predictors of perceived safety.

C. Existing Safety Applications

Table I compares existing applications. Most focus on reactive measures such as panic buttons, require internet connectivity, and lack offline or community-based features [4], [7], [10]. Prior work on SafeStreet [4] and Navi.io [7] shows the value of privacy-aware and crowd-sourced approaches. However, neither integrates offline SMS fallback, evidence capture, or fake-call diversion. Anoop and Bargavi proposed a Flutter-based safety app with SOS, GPS, and self-defense tutorials, but it remains reactive [2]. Baufeldt and Vanderschuren investigated ride-share users' safety perceptions in Cape Town and found gender-based differences before, during, and after Covid-19 [5].

TABLE II: Participant demographics

Characteristic	Distribution
Gender	100% female
Age range	21–25 (14/16), 26–35 (2/16)
Occupation	Students (14/16), working professionals (2/16)
Primary transport	Ride-sharing (Uber, Pathao), buses, CNGs
Harassment experience	14 of 16 reported direct or witnessed incidents

TABLE III: Study overview

Method	Participants	Output
Survey	16	Risk & feature counts
Interviews	2	Lived experiences
Affinity diagram	16 responses	6 themes
Prototype	5 features	Hi-fi screens

D. HCI and Feminist Design for Safety

Feminist HCI emphasizes participation, pluralism, advocacy, and ecology in designing gender-sensitive technologies [3], [8]. Participation means involving women as co-designers rather than treating them as passive users. Pluralism means acknowledging that women's needs differ by age, income, education, and location. Advocacy means designing for the empowerment of women, not just for their protection. Ecology means considering the broader social and legal context in which technology is used. Our work builds on these principles by involving women as co-designers and offering multiple activation methods, customizable trusted-contact lists, and privacy-preserving anonymous reporting.

III. Participants Overview

We recruited 16 female ride-sharing users for the survey and 2 additional users for in-depth interviews. All participants resided in Dhaka and were regular users of ride-sharing services such as Uber, Pathao, and inDrive. The sample was young adult/student-heavy (14 out of 16 survey respondents were aged 21–25, 14 were students). Ten had used Uber in the last 12 months; seven had used Pathao. Table II summarizes the demographic characteristics.

The sample is strongest for young adult and student experiences. Future work should broaden recruitment across age, income, occupation, and route diversity. Participants were recruited through university email lists and social media groups. Interviews were conducted in Bengali, the native language of the participants, which helped build rapport.

IV. Applied Research Methods

We used a mixed-methods approach combining a quantitative survey, qualitative interviews, and affinity diagramming. Table III summarizes the study.

A. Survey

We distributed an anonymous digital survey with closed-ended and rating questions. Sixteen female ride-sharing users

TABLE IV: Top safety concerns reported (N=16)

Concern	Count
Driver behavior/comments	7
Isolated or unfamiliar road	6
Unexpected route change	5
Travelling alone at night	4

TABLE V: Most requested safety features (N=16)

Feature	Count
One-tap live location	13
SOS to chosen contact	7
Verified driver & vehicle	6
Unexpected route alert	4

responded. The survey covered: safety concerns, pre-ride verification habits, location-sharing behavior, emergency feature preferences, awareness of existing safety features, and privacy comfort.

B. Interviews

We conducted two in-depth semi-structured interviews with female ride-sharing users in Dhaka. Interviews lasted 30–45 minutes and were conducted in Bengali. The protocol covered background and daily travel, planning, safety experiences, current methods, ride-sharing app use, emergency help, SOS preferences, location sharing, route changes, privacy, connectivity, reporting, and top feature priorities.

C. Affinity Diagramming

Survey responses and interview notes were grouped into affinity clusters. Six themes emerged: (1) Safety Concerns, (2) Verification, (3) Trusted Support, (4) Emergency Alerts, (5) Awareness Gap, and (6) Privacy & Trust.

D. Data Analysis

Survey counts were tallied for each concern and feature. Interview transcripts were thematically coded. Initial codes were grouped into the six affinity themes, which were then translated into design requirements.

V. Findings

A. Survey Results

1) Top Safety Concerns: Participants ranked the situations that made them feel unsafe (Table IV). Driver behaviour and comments came first, followed by isolated or unfamiliar roads, unexpected route changes, and travelling alone at night.

2) Most Requested Safety Features: Participants selected the features they found most useful (Table V).

3) Pre-Ride Verification Behavior: Seven participants check driver name/profile, seven check vehicle/number plate, five check ratings/reviews, and six selected a verified-driver feature as important. Verification is already part of the routine for many women, but the tools they use are informal rather than integrated into the ride flow.

4) Trusted Support Behavior: Seven share live location to feel safer, seven said their first action in danger is to share live location, and five call or message a trusted person first.

5) Awareness Gap: Four were not aware of existing safety features, five had never used one, and five said finding safety features was difficult or very difficult. Only two said it was easy.

6) Privacy and Trust: Eight rated identity privacy as important or very important. Comfort with sharing data with the app or police was mixed; participants were more comfortable sharing with a chosen contact.

B. Interview Themes and Quotes

1) Fear of Crowded Spaces and Harassment: One participant described the discomfort of crowded buses:

> "*The bus is very crowded. You face different awkward situations, like people pushing each other, people climbing on top of each other, falling when the bus brakes.*"

She added:

> "*Most of the time I feel unsafe and uncomfortable when traveling by bus.*"

Another participant described her response when she felt unsafe:

> "*Stand strong, did not appear weak. Made sure I was in a crowded place and didn't engage much with a stalker.*"

2) Coping Strategies: A participant carried protection and sat strategically:

> "*I carry a knife, try to sit on the front seats of a local bus if I am travelling by them. If I am on Uber, my family or friends—at least one person knows and tracks my activity.*"

3) Emergency Response Preferences: One participant described an active response:

> "*If I am in a car I will try to strangle the driver from the back seat. Otherwise stay calm and call someone and make sure the driver listens that on the location someone is waiting for me, they know I am on the way, and make sure to contact anyone who's near me.*"

4) Automatic Location Sharing: Both participants wanted automatic sharing. One described what she wanted shared:

> "*Live share—which road I'm taking, which alley I'm entering, road info, live location share.*"

5) Privacy Concerns: One participant emphasized anonymous reporting:

> "*Publicly my name should not show, from the platform. I'll report but my profile info should not be shared publicly.*"

6) Connectivity and Offline Functionality: One participant highlighted offline functionality:

> "*Battery and network problem so make sure it works offline.*"

TABLE VI: Affinity themes and design direction

Theme	Design Direction
Safety Concerns	Surface risk states (driver behavior, isolated road, night)
Verification	Show driver and vehicle details before ride start
Trusted Support	One-tap live location to chosen contacts
Emergency Alerts	Visible SOS and route deviation alerts
Awareness Gap	Place controls on active ride screen, not menus
Privacy & Trust	Anonymous reporting, identity hiding

TABLE VII: Task-based evaluation results

Task	Success	Avg. Time (s)
Account setup	100%	45
Trusted contacts	100%	38
SOS gesture	90%	52
Vehicle check	100%	60
Fake call	100%	15
Anonymous report	90%	40
Community map	80%	25

7) Most Important Features: One participant listed her top feature:

> "*If I don't reach within a certain time, auto alert goes to trusted person.*"

C. Affinity Themes

Table VI shows the six affinity themes and the design direction each provides.

D. Task-Based Evaluation

We conducted a task-based evaluation with the design team and target users. Participants completed seven tasks: (1) account setup, (2) adding trusted contacts, (3) configuring SOS gesture, (4) performing a vehicle safety check, (5) using the fake call, (6) submitting an anonymous report, and (7) viewing the community map. Table VII presents the results.

Participants praised the one-tap live location, the visible SOS button, and the offline SMS fallback. Criticisms included battery drain from continuous GPS, occasional false triggers from shake detection, and the need for cloud backup options for evidence. The 80% success rate on the community map task was the lowest, and participants noted that the map was crowded when many reports were visible. Average ratings were 4.4 for perceived safety, 4.6 for perceived usefulness, and 4.1 for privacy comfort.

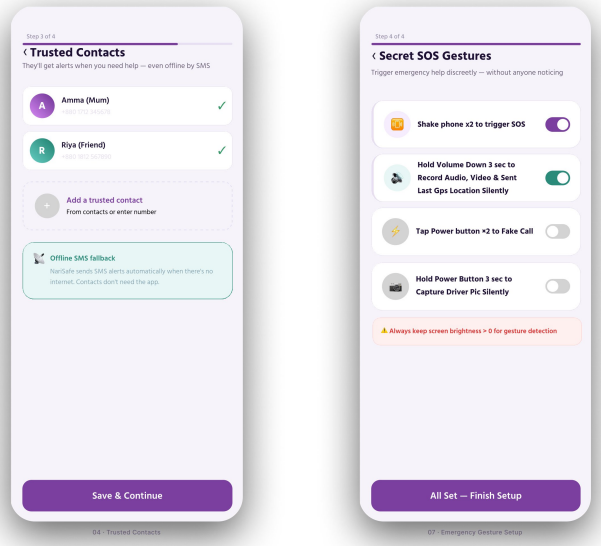
VI. Prototype Design

Based on the survey, interviews, and affinity themes, we designed NariSafe with five core features: (1) One-Tap Live Location, (2) Verified Driver & Vehicle Card, (3) Unexpected Route Alert, (4) SOS to Trusted Contact, and (5) Private Incident Reporting. The prototype also includes three supporting features: a recording mode that captures evidence during a trip, a fake call feature that lets the rider exit uncomfortable situations gracefully, and an offline SMS fallback that ensures alerts reach trusted contacts even when internet is unavailable.

A. Design Rationale and Interface Screens

NariSafe is a high-fidelity mobile prototype. Figures 1–7 show all thirteen screens of the application, grouped into seven figure blocks by function. The interface uses a warm purple-and-teal palette that avoids the aggressive red-and-yellow look of typical panic-button apps. Purple signals calm and dignity; teal signals reassurance and safety. Red is reserved only for true emergency actions. Bengali and English are both supported and can be switched from the setup screen.

The first screen block (Figure 1) shows the setup flow. The trusted contacts screen uses a clean card layout with avatar circles and check marks, mirroring familiar messaging apps. An "Offline SMS fallback" banner confirms that these contacts will receive SMS alerts even when the user has no internet. This is one of the most important design decisions in the prototype: a participant explicitly said "make sure it works offline," and this banner makes that promise visible from the very first setup step. The secret gestures screen offers four activation methods—shake x2, hold volume down for 3 seconds, tap power button twice, and hold power button for 3 seconds—responding to the feminist HCI value of pluralism. Not every woman will prefer a visible button, and not every woman will prefer a gesture; supporting both refuses a one-size-fits-all solution.

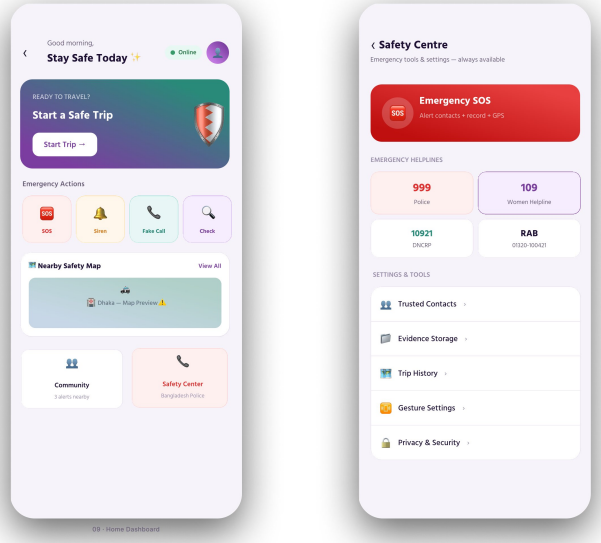


(a) Trusted Contacts (b) Secret SOS Gestures

Fig. 1: Setup screens: (a) Trusted Contacts lets the rider save contacts who receive alerts even offline by SMS. (b) Secret SOS Gestures configures four discreet activation methods.

The second screen block (Figure 2) shows the home and safety centre. The home dashboard places the most urgent actions without scrolling—four emergency tiles (SOS, Siren, Fake Call, Check) below the trip start button, followed by a nearby safety map preview, and two larger cards linking to the Community feed and the Safety Centre. The Safety Centre is

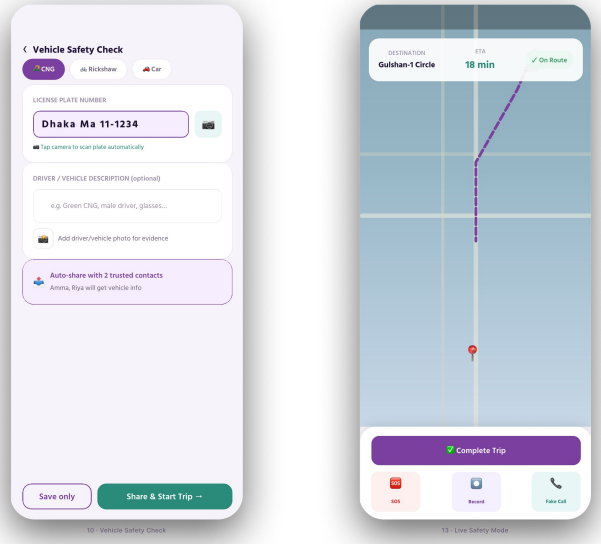
the second most important screen after the trip view. At the top sits a large red “Emergency SOS” tile that triggers an alert with a single tap and begins recording. Below that are four emergency helplines: 999 for police, 109 for the Women Helpline, 10921 for the DNCRP, and a dedicated RAB number. Listing these numbers inside the app solves a very practical problem: in an actual emergency, women should not have to remember the correct numbers or search for them in a browser. The lower half of the screen shows a vertical list of settings: Trusted Contacts, Evidence Storage, Trip History, Gesture Settings, and Privacy & Security.



(a) Home Dashboard (b) Safety Centre

Fig. 2: Home and safety centre: (a) Home Dashboard shows “Start a Safe Trip”, emergency action tiles, nearby safety map, and the Safety Centre entry. (b) Safety Centre provides one-tap Emergency SOS, emergency helplines (999, 109, 10921, RAB), and settings.

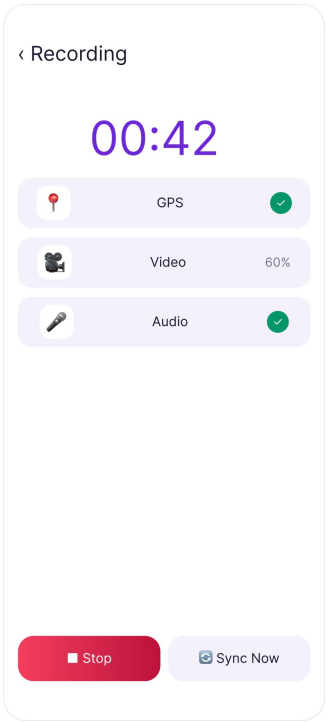
The third screen block (Figure 3) shows the trip screens. The vehicle safety check lets users select a vehicle type (CNG, Rickshaw, or Car) using three pill-shaped buttons at the top. Below, a text field labelled “License plate number” is pre-filled with an example format “Dhaka Ma 11-1234.” Next to the field is a camera icon that launches the plate scanner. At the bottom, a purple card confirms “Auto-share with 2 trusted contacts: Amma, Riya will get vehicle info.” This screen is placed before every trip so that the verification habit that seven out of sixteen participants already perform manually becomes a built-in step in the ride flow. The live safety mode screen combines route tracking with quick action buttons at the bottom. A card at the top shows destination, ETA, and a green “On Route” badge. At the bottom of the screen, four buttons sit in a horizontal row: Complete Trip, SOS, Record, and Fake Call. By keeping these four actions permanently visible during a trip, NariSafe makes it impossible to miss them.



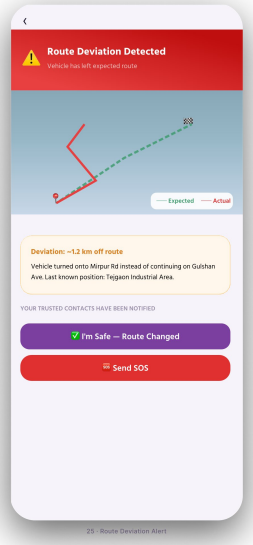
(a) Vehicle Safety Check (b) Live Safety Mode

Fig. 3: Trip screens: (a) Vehicle Safety Check records the license plate, driver description, and vehicle type, then auto-shares with trusted contacts. (b) Live Safety Mode shows the live route, ETA, and On-Route status.

The fourth screen block (Figure 4) shows recording and route alerts. The recording screen uses a calm, focused layout: a large timer at the top (00:42), then three status rows showing GPS, Video, and Audio. At the bottom sit two buttons: “Stop” (red, large) and “Sync Now” (light purple, small). The aim is to avoid escalating the user’s stress—the recording continues silently in the background while the user does something else. A small caption at the very bottom reads “Recording saved to encrypted local storage,” reassuring the user that the evidence is safe. The route deviation alert is one of the more novel screens. A red banner across the top reads “Route Deviation Detected: Vehicle has left expected route.” Below the banner, a split map shows the expected route as a dashed green line and the actual path as a solid red line. An amber card reports the deviation distance and last known position. A purple banner confirms that trusted contacts have already been notified. This screen is a direct response to a participant’s request for an emergency alert to the app handler team.

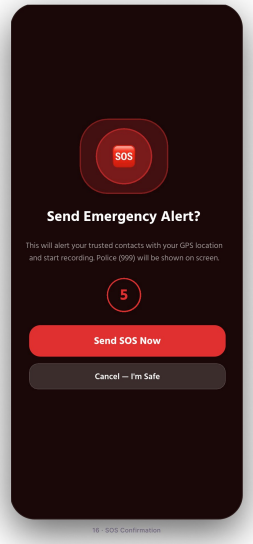


(a) Recording

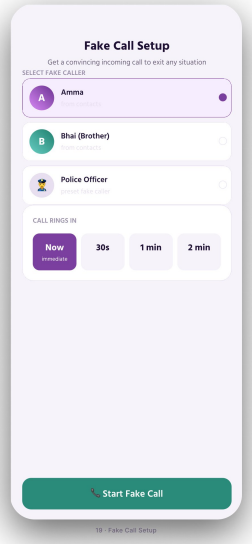


(b) Route Deviation Alert

Fig. 4: Recording and route alerts: (a) Recording screen shows a live timer with GPS, video, and audio status indicators. (b) Route Deviation Alert displays the expected vs. actual path and offers "I'm Safe" and "Send SOS" options.



(a) SOS Confirmation

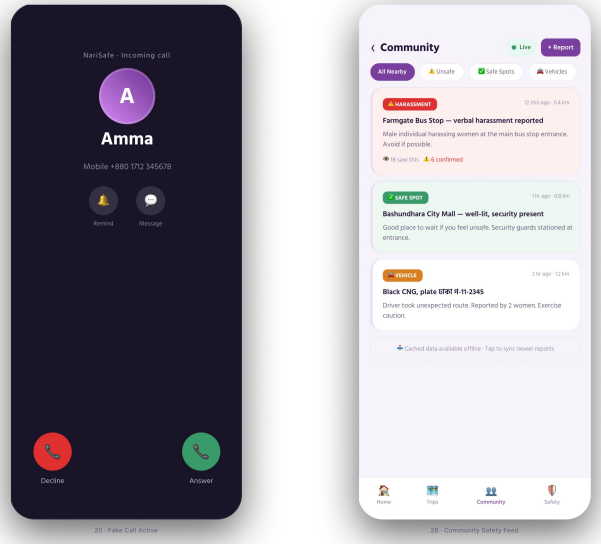


(b) Fake Call Setup

Fig. 5: Emergency screens (1): (a) SOS Confirmation displays a countdown and a large "Send SOS Now" button. (b) Fake Call Setup lets the rider choose a fake caller and set the ring delay.

The fifth screen block (Figure 5) shows the SOS confirmation and fake call setup. The SOS confirmation screen uses a full-screen dark background with a prominent red SOS icon in the middle and a 5-second countdown ring below. The heading reads "Send Emergency Alert?" and a caption explains that trusted contacts will be alerted with the GPS location and recording will begin. Two buttons sit at the bottom: "Send SOS Now" (solid red, large) and "Cancel — I'm Safe" (outlined grey, large). The countdown gives the user a chance to cancel an accidental trigger. Notably, the confirmation screen does not call the police automatically, reflecting the finding from Karusala and Kumar that Indian women distrust police response and prefer to alert family first [8]. NariSafe shows the police number so the user can call if she chooses, but does not make that choice for her. The fake call setup screen offers three caller options (Amma, Bhai, Police Officer) and four delay options (Now, 30s, 1 min, 2 min). This feature directly responds to a participant's account of feeling trapped in a car with no easy exit. Instead of physically confronting the driver, the user can gracefully exit by pretending to answer a phone call from her mother or brother.

The sixth screen block (Figure 6) shows the fake call active screen and the community safety feed. The fake call active screen is designed to look exactly like the phone's native incoming call screen. A small line at the top reads "NariSafe · Incoming call," followed by a large circular avatar with the letter "A" for Amma. Below the avatar is the caller name in large bold text and the mobile number in smaller grey text. Two small buttons (Remind, Message) sit mid-screen. At the bottom, a red decline button on the left and a green answer button on the right complete the familiar pattern. This exact mimicry is critical—any deviation from the native screen would reveal the ruse. The community safety feed uses colour-coded report cards: pink for harassment, pale green for safe spots, and pale orange for vehicle concerns. Each card shows a timestamp, distance, bold headline, short description, and a small footer that shows how many people saw and confirmed the report. Filters at the top let the user switch between categories (All Nearby, Unsafe, Safe Spots, Vehicles). A small note at the bottom of the screen reads "Cached data available offline · Tap to sync newer reports," which directly implements the offline-first design principle.



(a) Fake Call Active (b) Community Safety Feed

Fig. 6: Emergency screens (2): (a) Fake Call Active mimics a realistic incoming call from "Amma." (b) Community Safety Feed shows nearby reports with filters and an offline cache notice.

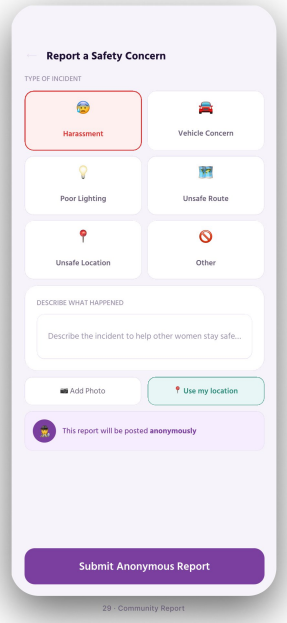


Fig. 7: Report a Safety Concern: the rider selects an incident type, adds a description, photo, and location, and submits an anonymous report.

The seventh screen block (Figure 7) shows the report a safety concern screen. It uses large, easy-to-tap incident tiles arranged in a two-by-three grid: Harassment, Vehicle Concern, Poor Lighting, Unsafe Route, Unsafe Location, and Other. The selected tile changes to a pink background with a bold red border, so the choice is visually confirmed. Below the tiles, a description field invites the user to "Describe the incident to help other women stay safe…" Two action buttons (Add Photo, Use My Location) sit above a purple banner that confirms the report will be posted anonymously. A large solid purple button at the bottom reads "Submit Anonymous Report." After submission, a confirmation screen returns a reference number like "NS-0929-4417" so that the user can follow up if needed. This anonymity is essential. As one participant said, "Publicly my name should not show, from the platform." In a society where victim-blaming is common, women will not report if they fear exposure, and without reports the community safety map cannot function.

VII. Discussions

Visibility beats feature quantity. Safety controls must be placed on the active ride screen, not hidden inside menus. Five out of 16 participants said finding safety features was difficult, and one explicitly asked for an on-screen option to share location instantly. Safety app designers should prioritize placement over volume.

Trust is social, not only platform-based. Participants felt safer when chosen family or friends received trip status, route alerts, and SOS signals. Eight rated identity privacy as important, and one said her name should not appear publicly. Our results show that in the Global South, the first point of contact should be the family, not the state.

Proactive flow is the design gap. Verification, route monitoring, and late-arrival signals help riders act before a situation becomes critical. One participant's top feature was an auto alert to a trusted person if she did not reach her destination on time—a proactive, not reactive, mechanism. This directly informs NariSafe's route deviation alert.

Offline-first is non-negotiable. Both interview participants explicitly mentioned network and battery constraints. Design decisions in NariSafe, such as SMS fallback and offline caching of community reports, reflect this. Without offline support, safety apps are unreliable precisely when they are needed most.

Anonymity enables community intelligence. One participant's insistence that her name should not appear publicly is a strong signal that anonymity is not a nice-to-have but a fundamental requirement for community reporting. In a society

where victim-blaming is common, women will not report if they fear exposure.

Design implications. For ride-sharing platforms: integrate verified driver cards and route alerts directly into the active ride view. For policymakers: support offline SMS fallback and community safety reporting. For designers: prioritize privacy, anonymity, and cultural sensitivity in the Global South context. For researchers: continue to study how safety perception changes across age, income, and region.

VIII. Conclusion, Limitations, and Future Work

We presented NariSafe, an HCI-driven prototype for women's ride-sharing safety in Dhaka. Through a survey (N=16), two interviews, and affinity diagramming, we identified six themes and translated them into five prototype features: one-tap live location, verified driver and vehicle card, unexpected route alert, SOS to trusted contact, and private incident reporting. The prototype was evaluated through task-based testing and found to have high usability, though battery life and evidence storage remain open challenges.

Limitations. The survey sample is small and skewed toward young adult students. Interview data comes from only two participants. The prototype was evaluated for usability, not for field effectiveness. Results should be treated as design evidence, not as citywide statistical generalization.

Future Work. We plan to (1) broaden recruitment across age, income, and occupation; (2) deploy NariSafe in a field study with ride-sharing platforms; (3) add AI-based threat detection for voice and motion signals; (4) integrate community reporting with police response systems; and (5) study long-term effects on perceived safety and trust. We also plan to translate the prototype into a functional Android app and explore partnerships with local ride-sharing companies.

References

[1] M. A. Abdal, R. M. Rafsanzani, and M. S. Rahman, "Assessment of Women's Safety in Ride-Sharing Services of Dhaka City," Bachelor's thesis, Islamic University of Technology, Gazipur, Bangladesh, 2022.

[2] I. Anoop and S. K. M. Bargavi, "Design and Development of a Women's Safety Mobile Application Using Flutter," *Journal of Advanced Scientific Computing*, vol. 15, no. 4, 2026.

[3] A. Z. Asha, S. Sultana, H. A. He, and E. Sharlin, "'Shotitwo First!': Unraveling Global South Women's Challenges in Public Transport," in *Proc. DIS '24*, Copenhagen, Denmark, 2024, pp. 1–17.

[4] M. E. Ali, S. B. Rishta, L. Ansari, T. Hashem, and A. I. Khan, "SafeStreet: Empowering Women Against Street Harassment using a Privacy-Aware Location Based Application," in *Proc. ICTD '15*, Singapore, 2015, pp. 1–4.

[5] J. L. Baufeldt and M. Vanderschuren, "Personal safety perceptions of ride-share users in Cape Town before, during and after Covid-19," *Transportation Research Interdisciplinary Perspectives*, vol. 20, 2023.

[6] S. Hu and Y. Yang, "Safety of female ride-hailing passengers: Perception and prevention," *Humanities and Social Sciences Communications*, vol. 11, no. 265, pp. 1–16, 2024.

[7] X. Y. Huang, E. C. West, and S. S. K. Pinnelli, "Crowdsourcing Data for Safer Travel Experiences for Women in India," in *Proc. CHI EA '23*, Hamburg, Germany, 2023, pp. 1–6.

[8] N. Karusala and N. Kumar, "Women's Safety in Public Spaces: Examining the Efficacy of Panic Buttons in New Delhi," in *Proc. CHI '17*, Denver, CO, USA, 2017, pp. 3340–3351.

[9] A. Meshram, P. Choudhary, and N. R. Velaga, "Assessing and Modelling Perceived Safety and Comfort of Women during Ridesharing," *Transportation Research Procedia*, vol. 48, pp. 2852–2869, 2020.

[10] S. Singh, T. Sanap, M. Singh, and K. Patil, "Women Safety Alert System: An Android Application for Real-Time Emergency Response," *IJARCCE*, vol. 15, no. 4, pp. 1865–1868, 2026.

[11] Y. Yang, S. Hu, D. Liao, and X. Huang, "What Affects Safety Perception of Female Ride-Hailing Passengers? An Empirical Study in China Context," *Journal of Advanced Transportation*, vol. 2022, Article ID 3316535, pp. 1–16, 2022.